DESIGN OF MODEL PREDICTIVE CONTROLLER (MPC) FOR MAGNETIC SUSPENSION SYSTEM

MUHAMMAD IBRAHIM ZUBAIR

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APPROVAL PAGE

This is to certify that this project titled "**Design of model predictive controller for magnetic suspension system**" was found to satisfy the requirements for the award of Bachelor of Engineering (B.Eng.) degree from the Department of Electrical Engineering K.U.S.T Wudil.

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_____

Engr.Dr Muhammad sani gaya
Date

(Project Supervisor)

_____

_____

Engr.Dr.Bashir Danjuma Bala.
Date

Head of Department

_____

_____

Prof.Ado Dan-Isa Date

External examiner

DEDICATION

This report is dedicated to my parent, May Allah forgive them, have mercy on them and guide them in all their endeavors.

ACKNOWLEDGEMENT

I wish to express my profound gratitude to Almighty Allah; my Creator, Guider, Sustainer, and Fashioner, with whose mercies and blessings I certainly exist. I would like to express my appreciation to the entire family of Alh. Ibrahim zubair , all my guardians from all perception and good wishers. Special thanks to my project supervisor, Engr.Dr.Muhammad sani gaya , who did not only supervise the project but also guided the work. Thank you for all the inspirational thoughts and the support you've given to me. May Almighty Allah reward you with the best of both worlds, Amin.

Thanks go to all my lecturers, who have taught me more than Electrical Engineering. While my journey only starts now, I will not forget what I have learnt from you all. Thank you all for your patience and guidance.

Thanks to all the technical staff, who provided the practical aspect of our study. May God in his abundance reward you all greatly, Amin.

ABSTRACT

Magnetic Levitation System (MLS) is highly dynamical process and open loop unstable system. The control of MLS using conventional controllers and methods are undesirable and unrealistic. Therefore this project is cantered on design of Model predictive controller (MPC) to obtain proper stability. Simulation result obtained has shown that MPC has better performance than PID controller. The proposed MPC is useful for the maglev system.

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LIST OF ABBREVIATIONS

MPC	Model Predictive Control
PID	Proportional Integral and Derivative

Chapter 1

INTRODUCTION

1.1 Introduction

Magnetic levitation system is a method to levitate object by using electromagnetic force only. The force from the magnetic field counteract with the gravitational force which make the object float. Magnetic levitation is not a new thing in engineering in fact Robert Goddard and Emile Bachelet in (1990's) is the earliest people that seen the theory of magnetic levitation[2]. Maglev technology has been receiving increasing attention due to its ability to eliminate energy losses due to Friction. Centred on friction reduction, maglev systems have wide engineering Applications such as magnetic bearings, high-precision positioning platforms, Aerospace shuttles, and fast maglev trains [3].

The object of this project is to keep a metal ball suspended in mid-air by adjusting the field strength of an electromagnet. The electromagnet current may be increased until the Magnetic force produced is equal to or greater than, the gravitational force acting on the ball. Variations in the electromagnet current cause the ball to either fall (when current is decreasing) or are attached to the electromagnet (when current is increasing). The MPC introduced aims to stabilize the ball when current disturbance occurs.

From the control engineering point of view, an MLS (or Maglev as it is sometimes called) is a quite complex system since it presents non-linearity and it is naturally unstable. The easiest (but not the best) way to handle this system is to consider only small variations around a given operating point. This allows linear control principles to be applied [4].

1.2 Problem statement

Improving equilibrium point between generated upward electromagnetic force and gravitational attraction force, which result in stabilized suspended iron ball

1.3 Aims and Objectives

The aim is to design MPC controller for magnetic levitation system.

The objectives are:

1. To stabilize a non linear unstable magnetic levitation system
2. To design MPC controller
3. To Compare the performance of MPC with PID

1.4 Scope of the research

The project only focused on designing MPC controller for Magnetic levitation system.

1.5 Project outline

Chapter two is the literature review which discusses the modelling and control of the system. Chapter three describes the methodology of the project. Chapter four presents analysis and simulation results of the project. And finally, the last chapter which is chapter five gives conclusion, recommendation, reference

Chapter 2

LITERATURE REVIEW

2.1 Introduction

This chapter describes review on magnetic suspension system, modelling of magnetic levitation and control techniques

2.2 Magnetic levitation system

Magnetic levitation is a method by which an object is suspended with no supports other than magnetic fields. Magnetic force is used to counteract the effects of the gravitational acceleration and any other acceleration.

The two primary issues involve in magnetic levitation are lifting force; providing an upward force sufficient to counteract gravity, and stability; ensuring that the system does not spontaneously slide or flip into a configuration where the lift is neutralized.

MLS are generally highly nonlinear and open loop unstable systems. This unstable aspect of MLS and its inherent nonlinearities make the modeling and control problems very challenging. Several dynamic models of magnetic force have been proposed over the past years and with these models various control strategies have been used comparing their performance. Both the linear and nonlinear techniques have been used. Linear system model only works well over a small region of operating point [4].

Wong obtained an approximate linear model, with an open-loop pole in the right-half plane. A phase-lead (linear) compensator was used to stabilize the system for step responses of 1.5 mm around the operating point [5]. Guess and Alciatore examined the differences between the conventional magnetic levitation system model and actual system. Effects of un-modeled dynamics on the stability of a simulated system were also studied. PID controller proved to be effective for set point regulation and for tracking a changing input [6]. Valer and Lia build a nonlinear model for magnetic levitation system and proposes systems linearization principle (the expansion in Fourier series and the preservation of the first order terms) in order to linearize the acquired nonlinear model [7].

Ying-Shing Shiao, (2001) employed system linearization and phase-lead compensation with virtual pole cancellation to design the controller of unstable nonlinear system to maintain better stability in a levitated ball. Such magnetic levitation systems (MLS) with small operating ranges have been proposed by the various researchers [8].

[9] presented a nonlinear model for the magnetic force of magnetic levitation device and model was then used to propose a control technique for position control of a magnetically levitated permanent magnet. A Lyapunov based stability analysis was performed to prove the stability of the control technique. It was reported that the proposed controller performed a precise positioning operation over an operation range of 30 mm, which is an improvement over available control strategies in the literature for large gap magnetic levitation systems.

In [10] the author carried out a comparative study of linear and nonlinear controllers for Maglev system and stated that, feed-back linearization controller has provided significantly better trajectory tracking.

METHODS TO MAKE MAGNETIC LEVITATION SYSTEM

There are several ways to levitate an object. Among them is servo-stabilized electromagnetic suspension (EMS), electrodynamics suspension (EDS).

- SERVO-STABILIZED ELECTROMAGNETIC SUSPENSION (EMS)

The attraction from a fixed strength magnet decreases with increase distance, and increases at closer distance. This is unstable. For a stable system, the opposite is needed; variation from a stable position should push it back to the target position.

Stable magnetic levitation can be achieve by measuring the position and speed of the object being levitated, and using a feedback loop which continuously adjust one or more electromagnet to correct the objects motion, thus forming servomechanisms.

- ELECTRODYNAMICS SUSPENSION (EDS)

In the electrodynamics suspension system, the object is lift by the magnetic forces that Act on the object and the base. The magnetic field is produced by either

electromagnets or by an array of permanent magnets. The repulsive force in the system is created by an induced magnetic Field in wires or other conducting strips in the track. EDS cannot occur at static it need some speed to gain the dynamic force. At slow speeds, the current induced in these coils and the resultant magnetic flux is not large enough to support heavy weight object. For this reason the object must have wheels or some other form of landing gear to support the object until it reaches a speed that can sustain levitation. Propulsion coils on the base are used to exert a force on the magnets in the object and Make the object move forward. The propulsion coils that exert a force on the object are effectively a linear motor: An alternating current flowing through the coils generates a continuously varying magnetic field that moves forward along the track. The frequency of the alternating current is synchronized to match the speed of the train. The offset between the field exerted by magnets on the train and the applied field create a force moving the train forward. Since the electromagnetic field is developed as the vehicle moves, the flux produced by the onboard coils induces currents produce a magnetic sheets on the guide way. As a result, the induced currents produce a magnetic flux that opposes the magnetic flux of the onboard electromagnet, producing repulsive forces between the object and the guide way. Since the repulsive forces are produce as the object move above the passive coil on the guide way, the vehicles cannot be lifted unless a certain speed is achieve and therefore the electrodynamics suspension system requires[1].

APPLICATION OF MAGNETIC LEVITATION

Magnetic levitation_ systems have advantages of friction less movement, isolation of Environment and high precision. Some of the applications of magnetic levitation are stated Below:

- MAGNETIC LEVITATION TRAIN:

Magnetic Levitation (Maglev) train has been the most important usages of magnetic levitation technology. The train moves along the guide way with the help of magnetic field which helps The train to levitate and propel. In March 2004 first Maglev commercially implemented was Sanghai‟s TransraPID system which uses German model and in April 2005 Japan implemented its own HSST „Limino‟.at relatively slow speed than Shanghai one. The fastest train „Lo‟ Japan Is the fastest train till recorded of speed 603km/hr.[11]

- MAGNETIC BEARING:

Magnetic bearing use rotor to levitate and rotate with magnetic flux interaction from stator mounted electromagnet. Since no contact thus no friction, no drag and no wear and tear of parts. Magnetic bearing are used in flywheel as energy storage device, as blood pump, micro positioning system and semiconductor industries.[11]

- LAUNCHING ROCKET:

NASA‟s Marshall Space Flight Center at Huntsvhill, Alabama has developed a track to magnetically levitate a space craft and give initial velocity to reach escape velocity of earth. The Project is to make space transportation with less cost. The space craft will be able to have 964km/hr when launched from the track without having any fuel consumption.[11]

2.3 Modelling and state space representation of the plant

The main challenge in controlling the position of the suspended object is to maintain the balance of the weight of the suspended object and the electromagnetic force acting on it. In this paper we are interested only in controlling the *vertical position* of the suspended object. Thus all the dynamics will be concerned to vertical motion of the suspended object [12].

The dynamics of the electromagnet is given by:

$$\frac{di}{dt} = \frac{1}{L(x)}\left[-Ri + L_0 a \frac{i}{(a+x)}\frac{dx}{dt} + v\right] \dots\dots\dots\dots\dots (2.1)$$

Where

V=the source voltage
r= series resistance of the circuit
i=current flowing in the electromagnetic

$$L(x) = L_1 + \frac{L_0}{1+\frac{x}{a}} \dots\dots\dots\dots\dots (2.2)$$

L_1=inductance when ball is not present
L_0=inductance when ball is next to coil
a=constant
The dynamics of suspended object in vertical plane is given by

$$m\frac{d^2x}{dt^2} = mg - k\dot{x} - F(x,i) \dots\dots. (2.3)$$

Where,

M = mass of the suspended object
G = acceleration due to gravity
x= distance of ball surface from the bottom of the electromagnet
k=coefficient of viscous friction
i=current in the coil
$f(x,i)$=force generated by the electromagnet
Now energy stored in the electromagnet is given as

$$\boldsymbol{E(x,i) = \frac{1}{2}L(x)i^2} \dots\dots\dots\dots (\mathbf{2.4})$$

Thus,the force is calculated as $f(x,i) = -\frac{\partial \mathrm{E}}{\partial \mathrm{X}} = \frac{L_0 a}{2}\frac{\mathrm{i}^2}{(\mathrm{x+a})^2}$ …… (2.5)

Using the above expression of $f(x,i)$, the differential equation governing the motion of the suspended object is given as

$$\frac{d^2x}{dt^2} = g - \frac{k}{m}\dot{x} - \frac{L_0 a}{2m}\frac{i^2}{(a+x)^2} \ldots\ldots\ldots (2.6)$$

Defining the state variable as

$x_1 = x$(Position)
$x_2 = \dot{x}$(Velocity)
$x_3 = i$(Current)
Control input as $u = v$
Output as $y = x_1$

The state model is given as:

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = g - \frac{k}{m}x_2 - \frac{L_O a}{2m}\frac{x_3^2}{(a+x_1)^2}$$
$$\dot{x}_3 = \frac{1}{L(x_1)}\left[-Rx_3 + L_0 a\frac{x_2 x_3}{(a+x_1)^2} + u\right]$$
$$y = x_1$$

Above nonlinear state model can be written as

$$\dot{x} = f(x) + g(x)u$$
$$y = h(x)$$

Where

$$f(x) = \begin{bmatrix} x_2 \\ g - \frac{k}{m}x_2 - \frac{L_0 a}{2m}\frac{x_3^2}{(a+x_1)^2} \\ \frac{-Rx_3 + L_O a\frac{x_2 x_3}{(a+x_1)^2}}{L_1 + \frac{L_0}{1+\frac{x_1}{a}}} \end{bmatrix}, \quad g(x) = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_1 + \frac{L_0}{1+\frac{x_1}{a}}} \end{bmatrix}, \quad h(x) = x_1$$

2.4 Linearization

The nonlinear dynamics of the system are linearized about the equilibrium point by Jacobi linearization and the linearized system is obtained as below:

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

With matrix

$$A = \left[\frac{\partial F(x,u)}{\partial x \partial}\right]_{(x^6,u^6)} = \begin{bmatrix} \frac{\partial F_1(x,u)}{\partial x_1} & \frac{\partial F_1(x,u)}{\partial x_2} & \frac{\partial F_1(x,u)}{\partial x_3} \\ \frac{\partial F_2(x,u)}{\partial x_1} & \frac{\partial F_2(x,u)}{\partial x_2} & \frac{\partial F_2(x,u)}{\partial x_3} \\ \frac{\partial F_3(x,u)}{\partial x_1} & \frac{\partial F_3(x,u)}{\partial x_2} & \frac{\partial F_3(x,u)}{\partial x_3} \end{bmatrix}_{(x^6,u^6)}$$

$$B = \left[\frac{\partial F(x,u)}{\partial u}\right]_{(x^6,u^6)} = \begin{bmatrix} \frac{\partial F_1(x,u)}{\partial u} \\ \frac{\partial F_2(x,u)}{\partial u} \\ \frac{\partial F_3(x,u)}{\partial u} \end{bmatrix}_{(x^6,u^6)} \qquad C = \left[\frac{\partial h(x)}{\partial x}\right]_{(x^6)} \quad \text{And} \quad D = \left[\frac{\partial h(x)}{\partial u}\right]_{(x^6)}$$

Where

$$F(x,u) = [F_1(x,u)F_2(x,u)F_3(x,u)]^T = f(x) + g(x)u$$

2.5 Control strategy

PROPORTIONAL – INTEGRAL – DERIVATIVE (PID) CONTROL

A proportional–integral–derivative controller (PID controller) is a controller which is popularly used in industrial control systems. It is fed with the error signal, that is, the difference between the reference, or the desired output and the actual output (which is obtained as a feedback). The controller then attempts to bring the actual output to track the reference. The structure of PID controller is showed in fig 1.

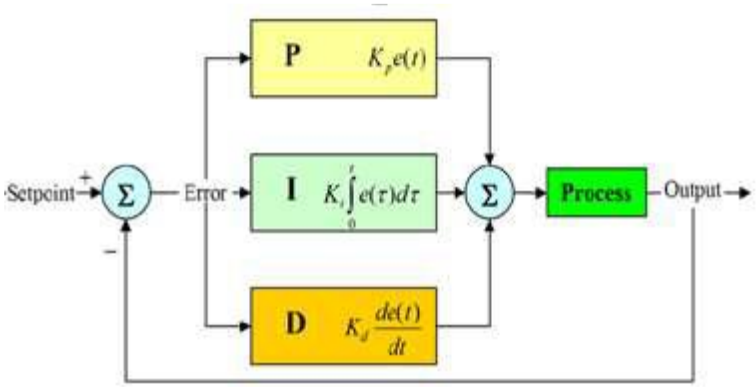


Figure 3.1 : Structure of PID Control (adopted from [13])

The PID controller algorithm involves three separate constant parameters (proportional, integral and derivative, denoted by P, I and D respectively) and is

thus, also called three- term control. P depends on the present error, I on the accumulation of past errors, and D is a prediction of future errors, based on current rate of change. The weighted sum of these three actions is given as input to the process through a control element. By tuning the three parameters, the controller provides the required control action.

If C(t) represents the controller output, which is the manipulated variable (MV(t)), that is, the process input, then:

$$C(t) = MV(t) = Kpe(t) + KI\int_0^t e(t)dt + Kd\frac{d}{dt}e(t) + P_{out} \dots\dots (2.7)$$

Effect of proportional gain: If the proportional gain is too small, it results in a small output response for a large input error. This, results in a large steady-state error.
Effect of integral gain: The integral gain removes the residual steady state error that occurs with a pure proportional controller. However, a very high integral gain results in overshoot.
Effect of derivative gain: The derivative gain is used to reduce the overshoot caused by the

integral gain and improve the combined controller-process stability. But, it slows down the transient response of the system as well as, increases sensitivity of the system to noise. So, a lead-compensator is used as an approximation of the differentiator.

In order to calculate the output of the PID controller, the three terms are summed together, which can be expressed as formula (2.7) [14] [15]:

For the control process, better performance can be achieved by tuning the control loop, which is Adjusting the control parameters to satisfy the desired control response.

Therefore, tuning PID control parameters is a complicated process that we have to find an optimal way to arrange the values of the parameters for the control response. In thus thesis, Ziegler-Nichols is widely used.

MODEL PREDICTIVE CONTROL (MPC)

Model Predictive Control (MPC) is an advanced method of process control that has been in use in the process industries in chemical plants and oil refineries since the 1980s. MPC also known as Receding horizon control or Moving horizon control uses an explicit dynamic plant model to predict the effect of future reactions of the manipulated variables on the output and the control signal obtained by minimizing the cost function. This predication takes into account, constraints on both the inputs and outputs of the process. An optimal input sequence is calculated. The measurements are then sent back to the controller, and a new optimizing problem is solved. [16]

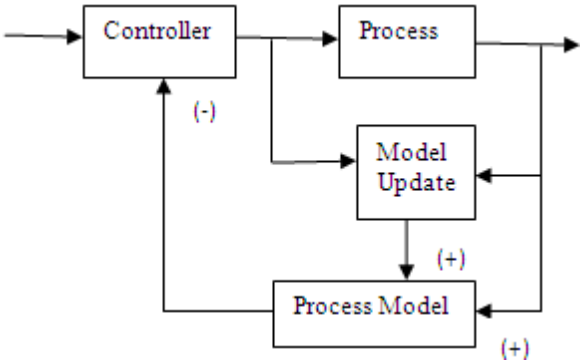


Figure 3.2 : Structure of MPC (adopted from [16])

2.6 Summary

This chapter has presented a literature study of magnetic suspension system. Different control strategies used for the system ware cited as the control of the system is inherently open-loop unstable. An accurate model of MLS is required to ensure the effective operation of the proposed model predictive control (MPC controller) .PID was discussed to give us a basis for comparison to the MPC later in the thesis.

Chapter 3

RESEARCH METHODOLOGY

3.1 Introduction

The chapter discusses the design of the MPC controller for the magnetic suspension system. The principle of operation is given followed by the discussion of the design algorithm used in *MATLAB® Model Predictive Control toolbox™* of the MPC controller. The design can be used for both SISO and MIMO system. Lastly, Procedure for the designing of MPC via SIMULINK is given.

3.2 Model predictive control strategy

Model predictive control (MPC) includes a class of control algorithms that utilize an explicit process model to predict the future response of a plant. At each control interval an MPC algorithm attempts to optimize future plant behavior by computing a sequence of future manipulated variable adjustments. The first input in the optimal sequence is then sent into the plant, and the entire calculation is repeated at subsequent control intervals. The following is a figure2 shows the basic idea of predictive control based on a single-input, single output plant.

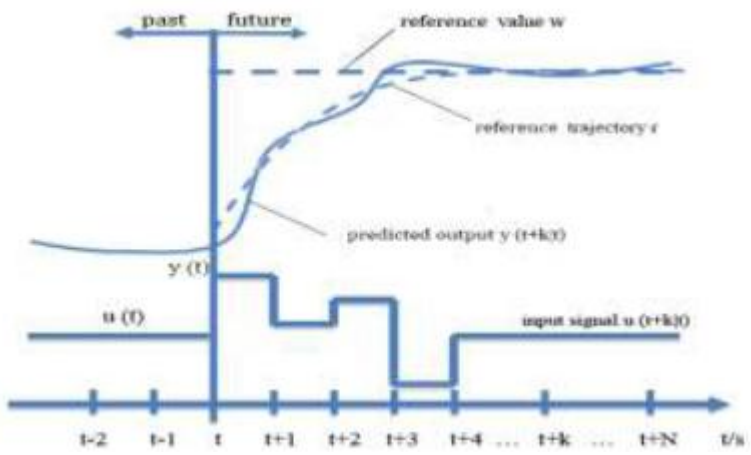


Figure 3.1: Model Predictive Control Strategy(adopted from [16])

We marked the current time as „t", with the plant output y(t). The figure also shows reference value *w*, reference trajectory „r" and control signal u(t+k | t).The period from „t" to „t+N" is called the prediction horizon, which determines the predicted output y(t+k | t) and dictates how „far" we wish the future to be predicted for. The objective of model predictive control law is to drive future plant outputs y(t+k | t) as close as *w* as shown in figure 3.1[16]

3.3 Matlab® model predictive control toolbox™

This section covers the theory of MPC controllers for MIMO systems, as used in the *MATLAB®Model Predictive Control toolbox™* (*MPC toolbox™*). It should be noted that this section will not discuss all of the functions of the *MPC toolbox™*; only the functions applicable to the design of the MPC controller for the magnetic suspension system will be discussed.

An advantage of using the *MPC toolbox™* is that MPC controllers can be designed and Implemented with relative ease. Also, with insight into the working of MPC, the parameters of the controller can be tuned to obtain a desirable response from the system [17]. The Development of the MPC controller using the *MPC toolbox™* is discussed according to the Model, observers, objective function and predictive control used in the design of the MPC Controller for the suspension system.

MODELS USED TO DEVELOP THE MPC CONTROLLER

The model of the magnetic suspension system discussed in section 2.2 is used to design the MPC controller. However, in order to improve the accuracy of the predictions, the effects of disturbances must also be included. This is done by including a model of the unmeasured disturbances and Measurement noise in the design.

The model of the magnetic suspension system can be expressed as a linear time-invariant model in state-space form

$$x(k+1) = Ax(t) + B_u u(k) B_v v(k) + B_d d(k)$$

$$y(k) = C_m x(k) + D_v v(k) + D_d d(k) \quad \ldots\ldots (3.1)$$

Where *x(k)* is an *nx*-dimensional vector of states, *u(k)* is an *nu*-dimensional vector of control variables, *v(k)* is an *nv*-dimensional vector of measured disturbances, *d(k)* is an *nd*-dimensional vector of unmeasured disturbances. *y(k)* represents an *ny*-dimensional vector of measured outputs. In (3.1) the unmeasured disturbance *d(k)* includes the state and output disturbances. These disturbances can be modeled as [17]:

$$x_d(k+1) = \bar{A}x(k) + \bar{B}n_d(k)$$

$$d(k) = \bar{C}x_d(k) + \bar{D}n_d(k) \quad \ldots\ldots (3.2)$$

Where *nd(k)* is assumed to be random zero-mean unit-variance white Gaussian noise. The plant Model is augmented with a model of the output disturbances in order to reject constant disturbances on measured outputs. This adds the integration effect to the design of the MPC controller.The final disturbance that can be included in the design is the effect of measurement noise *nm(k)*. The measurement noise can be modeled as the linear time-invariant (LTI) system driven by random zero-mean unit-variance white Gaussian noise as:

$$x_m(k+1) = \tilde{A}x_m(k) + \tilde{B}n_m(k)$$

$$m(k) = \tilde{C}x_m(k) + \tilde{D}n_m(k) \quad \ldots\ldots (3.3)$$

By appending the models of the disturbances and noise to the plant model an extended model Of the system is obtained. The immeasurable states *x(k), xd(k)* and *xm(k)* are obtained from the State estimator at each sampling instant.

OBSERVERS IN THE DESIGN OF THE MPC CONTROLLER

In order to reduce the prediction error the states *x(k)*, *xd(k)* and *xm(k)* are updated at each sampling instant by using a state observer. A condition for the use of state estimation is that the models used in the design of the MPC controller are observable. In order to check the observability the models are firstly converted to discrete-time models without delays. These models are then combined and numerically checked for observability [17].

Before the states of the system can be estimated, the output of the system must first be estimated from the known parameters as [17]:

$$\hat{y}(k) = C\hat{x}(k|k-1) + D_v v(k) + D_d \bar{C}\hat{x}_d(k|k-1) + \tilde{C}\hat{x}_m(k|k-1) \ldots\ldots (3.4)$$

As can be seen from (3.4) the estimated output *yˆ (k)* is calculated from the models discussed in the previous section, as well as the states estimated at the previous sampling instant (*xˆ (k|k-1)*).From *yˆ (k)* the states at the current sampling instant can then be calculated as [17]:

$$\begin{bmatrix} \hat{x}(k|k) \\ x_d(k|k) \\ \hat{x}_m(k|k) \end{bmatrix} = \begin{bmatrix} \hat{x}(k|k-1) \\ \hat{x}_d(k|k-1) \\ \hat{x}_m(k|k-1) \end{bmatrix} + K_{ob}\big(y(k) - \hat{y}(k)\big) \dots\dots (3.5)$$

where K_{ob} is the observer gain obtained from Kalman filtering techniques. This indicates that the current states are estimated from the states estimated at the previous sampling instant and a Correction term. The correction term updates the estimated states based on the error between The measured output *y(k)* and the estimated output *yˆ (k)*. The estimated states from (3.5) are used in the calculation of the optimal control.

The estimated states are also used to estimate the state at the next sampling instant as [17]:

$$\begin{bmatrix} \hat{x}(k+1|k) \\ \hat{x}_d(k+1|k) \\ \hat{x}_m(k+1|k) \end{bmatrix} = \begin{bmatrix} A\hat{x}|(k|k) + B_u u(k) + B_v v(k) + B_d \overline{C}\hat{x}_d(k|k) \\ \overline{A}\hat{x}_d(k|k) \\ \tilde{A}\hat{x}_m(k|k) \end{bmatrix} \dots\dots (3.6)$$

The estimates obtained from (3.6) are used at the next sampling period as the estimates of the previous states (*xˆ(k|k-1)*) as indicated in (3.5).

THE OBJECTIVE FUNCTION OF THE MPC CONTROLLER

Based on the estimates *x(k)*, *xd(k)* and *xm(k)* as calculated in (3.5), the optimal control action

uoptimal can be calculated. This is done by solving the optimization problem [17]

$$J = \min_{\Delta u(k|k),..,\Delta u(k|k+N_c-1)} \left[\sum_{i=0}^{N_p-1} \left[\sum_{j=1}^{n_y} \left| W_{i+1,j}^{y} \left(y_j(y+i+1) - r_j y_j(k+i+1) \right) \right|^2 + \sum_{j=1}^{n_u} \left| W_{i,j}^{\Delta u} \Delta u_j(k+1|k) \right|^2 \right] \right] \dots (3.7)$$

Subject to the constraint

$$U_j^{min}(i) \le u_j(k+1|k) \le u_j^{max}(i)$$

$$\Delta u_j^{min}(i) \le \Delta u_j(k+1|k) \le \Delta u_j^{max}(i)$$

$$y_j^{min}(i) \le y_j(k+1|k) \le y_j^{max}(i) \dots\dots (3.8)$$

with *i* = 0,1,2,T,*(Np*-1*)* and assuming that the increment in the control from the end of the control horizon to the end of the prediction horizon is $\Delta u(k+Nc|k)$,T,$\Delta u(k+Np|k) = 0$. This assumption is made in order to reduce the computational load.

As mentioned previously the use of linear inequality constraints with the quadratic cost function leads to a quadratic programming problem. The solution of the quadratic programming problem is the optimal control move *uoptimal*. In the unconstrained MPC case *uoptimal* is calculated analytically.

PREDICTIVE CONTROL DESIGN

The cost function given in (3.7) can be written as [17]:

$$J = \left\{ \begin{bmatrix} y(1) \\ \vdots \\ y(N_p - 1) \end{bmatrix} - \begin{bmatrix} r(1) \\ \vdots \\ r(N_p - 1) \end{bmatrix} \right\}^T W_y^2 \left\{ \begin{bmatrix} y(1) \\ \vdots \\ y(N_p - 1) \end{bmatrix} - \begin{bmatrix} r(1) \\ \vdots \\ r(N_p - 1) \end{bmatrix} \right\} + \begin{bmatrix} \Delta u(0) \\ \vdots \\ \Delta u(N_p - 1) \end{bmatrix}^T W_{\Delta u}^2 \begin{bmatrix} \Delta u(0) \\ \vdots \\ \Delta u(N_p - 1) \end{bmatrix} \dots . . (3.9)$$

With the weights *Wy* and *WΔu* representing the weights on the output and the incremental control, respectively. These weights are used to determine the speed and the robustness of the response. For larger values of *Wy* any deviations between the output and the set-point are penalized to a greater extent. This means that set-points are tracked more stringently. For larger values of *WΔu* greater care is taken when calculating the incremental control signal. This increases the robustness, but also has the effect that the response is slower.

By substituting *u(k), Δu(k)* and *y(k)* in (3.9) the cost function can be written in terms of the extended system model as [17]:

$$J = \Delta u^T K_{\Delta u} \Delta u + 2 \left\{ \begin{bmatrix} r(1) \\ \vdots \\ r(N_p) \end{bmatrix}^T K_r + \begin{bmatrix} v(0) \\ \vdots \\ v(N_p) \end{bmatrix}^T K_v + u(-1)^T K_u + x(0)^T K_x \right\} \Delta u + const \ (3.10)$$

In (3.10) $K_{\Delta u}$ represents the Hessian matrix. K_r, K_v, K_u and K_x are parameters relating the cost to the set-point over the prediction horizon, the measured disturbance over the prediction horizon, previous control actions and the initial states of the system, respectively.

In the case of unconstrained MPC the optimal incremental control action is calculated

Analytically using the following equation [17]:

In the case of constrained MPC the constraints are written in a similar form as (3.11)

$$\begin{bmatrix} M_u \\ M_{\Delta u} \\ M_y \end{bmatrix} \Delta U \leq \begin{bmatrix} N_u \\ N_{\Delta u} \\ N_y \end{bmatrix} \dots \dots \dots (3.11)$$

With

$M_u = \begin{bmatrix} -C_1 \\ C_2 \end{bmatrix}$; $N_u = \begin{bmatrix} -U^{min} + C_1 u(k-1) \\ U^{max} - C_1 u(k-1) \end{bmatrix}$; $M_{\Delta u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$; $N_{\Delta u} = \begin{bmatrix} -\Delta U^{min} \\ \Delta U^{max} \end{bmatrix}$;
$M_y = \begin{bmatrix} -\emptyset \\ \emptyset \end{bmatrix}$;

$$N_y = \begin{bmatrix} -Y^{min} + Fx(k) \\ Y^{min} - Fx(k) \end{bmatrix}$$

By relating the constraints on *u(k)*, *Δu(k)* and *y(k)* to the optimal control signal, the following Inequality is obtained [33]:

$$\Delta u_{optimal} = -K_{\Delta u}^{-1} \left\{ \begin{bmatrix} r(1) \\ \vdots \\ r(N_p) \end{bmatrix}^T K_r + \begin{bmatrix} v(0) \\ \vdots \\ v(N_p) \end{bmatrix}^T K_v + u(-1)^T K_u + x(0)^T K_x \right\}^T \quad \ldots (3.12)$$

From above the optimal control action $U_{optimal}$ can then be calculated as:

$$U_{optimal}(k) = u(k-1) + \Delta u_{optimal}(k) \ldots\ldots (3.13)$$

In the case of constrained MPC the constraints are written in a similar form as (3.11). By relating the constraints on *u(k)*, *Δu(k)* and *y(k)* to the optimal control signal, the following inequality is obtained [17]:

$$M_{\Delta u} \Delta u \leq M_{lim} + M_v \begin{bmatrix} v(0) \\ \vdots \\ v(N_p) \end{bmatrix} + M_u u(-1) + M_x x(0) \ldots\ldots (3.14)$$

The parameters of the inequality presented in (3.14) are then used in the quadratic programming problem to solve the optimal control action. The gains used to relate the constraints to the optimal control are $M_{\Delta u}$ that relates to the incremental control, M_{lim} that relates to the values of the constraints, M_v that relates to the modeled noise, M_u that relates to the past control actions, and M_x that relates to the past responses of the plant.

3.4 Design of mpc via matlab simulink

DEFINING INTERNAL PLANT MODELS

When designing a model predictive controller in SIMULINK, use SIMULINK Control Design to extract a linearized form of the SIMULINK model and automatically

import it into the controller as the internal plant model. [18][19] Alternatively, use of linear time-invariant (LTI) systems from Control System Toolbox, such as a transfer function or a state-space model, to specify the internal plant model. Then import LTI models from the MATLAB workspace or from MAT-files into the toolbox. The toolbox also lets us directly import models created from measured input-output data using System Identification Toolbox. [20]

DESIGNING CONTROLLER

Once the internal plant model is defined, complete the design of model predictive controller by specifying the following controller parameters:

Prediction and Control Horizons

Hard and Soft constraints on manipulated variables and output variables

Weight on manipulated variables and output variables

Models for measurement noise and for unmeasured input and output disturbances

In addition to constant constraints and weights, the toolbox supports time-varying constraints and weights, constraints on linear combinations of manipulated variables and output variables, terminal constraints and weights, and constraints in the form of linear off-diagonal weights. The toolbox also supports constraint softening. [20]

3.5 Summary

MPC controller has the ability to predict future action and take control action accordingly. The MPC tune parameters are known to be prediction horizon, sampling Time and control Horizon.MPC design algorithm of MATLAB® Model Predictive Control toolbox™ was discussed based on theories and more practical procedure in SIMULINK was designed.

Chapter 4

RESULTS AND DISCUSSION

4.1 INTRODUCTION

Simulation based on the mathematical model for nonlinear magnetic suspension system by using MATLAB/SIMULINK software has been performed. This gives greater possibility of testing and evaluating the capabilities of different control strategies which often very difficult and tedious.

4.2 simulation result

From linearization matrix obtained in chapter two above, substituting the parameters table below:

Table 4.1: Parameters table

Parameter	Symbol	Value
Suspended Object Mass	m	$0.1\,Kg$
Coil Inductance	L_1	$0.02\,H$
Incremental Inductance	L_0	$0.01\,H$
Coil Resistance	R	$1\,\Omega$
Gravity Acceleration	g	$9.81\,m/sec^2$
Coefficient of Viscosity	k	$0.001\,N/m.sec$
Constant 'a'	a	$0.05\,m$
Equilibrium Position	${x_1}^e$	$0.05\,m$
Initial Conditions	$[{x_1}^0 {x_2}^0 {x_3}^0]$	$[0.05 \;\; 0 \;\; 6.2642]$

We obtain the state space matrix as:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 196.2 & -0.01 & -3.1312 \\ 0 & 12.5284 & -40 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 40 \end{bmatrix} \quad C = [1 \quad 0 \quad 0] \text{ And } D = 0$$

Also transfer function of the system is obtain as:

$$G(s) = C[sI - A]^{-1}B + D$$

$$G(s) = \frac{-125.3}{s^3 + 40.01s^2 - 156.6s - 7848}$$

OPEN LOOP PERFORMANCE

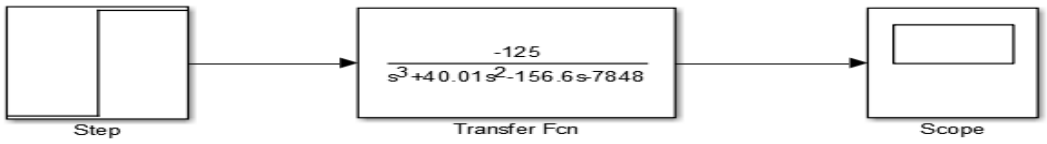


Figure 4.1 : open loop SIMULINK block due to step input

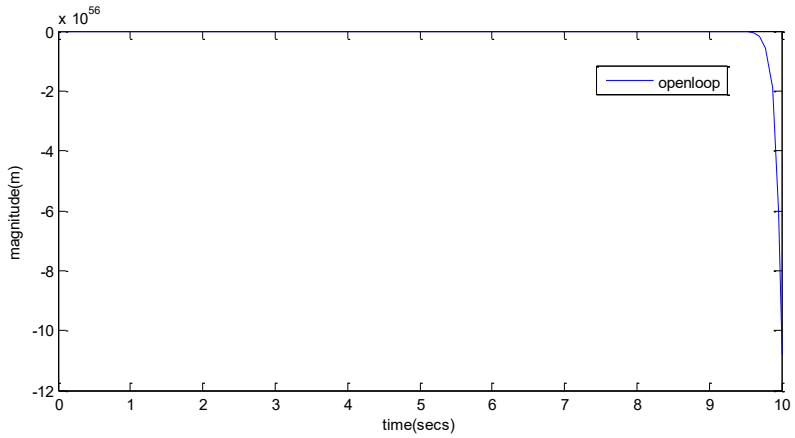


Figure 4.2 : Open loop response

PID PERFORMANCE

The response of the PID is as shown:

Table 4.2: PID tune parameters

INPUT	Proportional	Integral	Differential
Step	-152kj	-190	-16

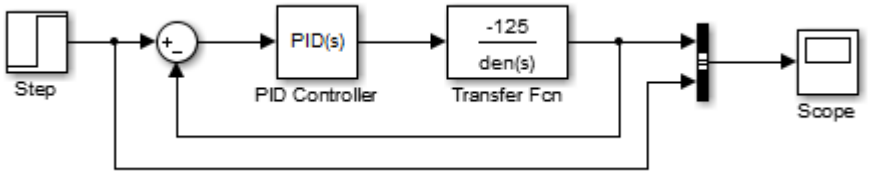


Figure 4.3 : PID SIMULINK block diagram

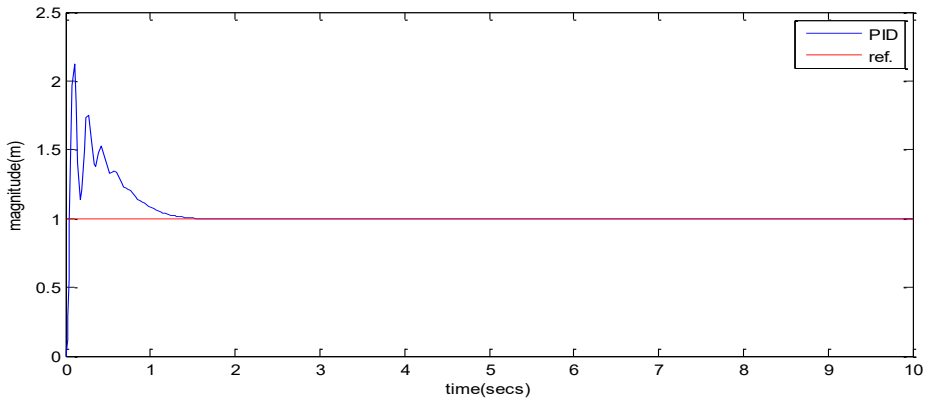


Figure 4.4 : PID response due to unit step input

MPC CONTROLLER PERFORMANCE

The MPC tune parameters and response of the system are as shown below:

Table 4.3: MPC tune parameters

INPUT	PREDICTION HORIZON	CONTROL HORIZON	SAMPLING TIME
Step	29	4	0.0064

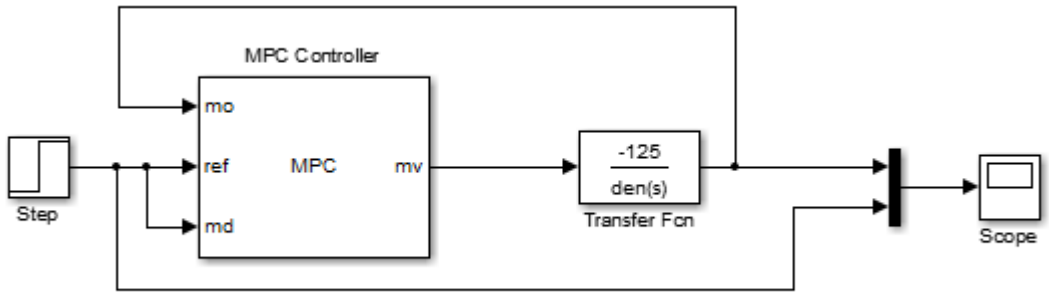


Figure 4.5 : MPC SIMULINK block

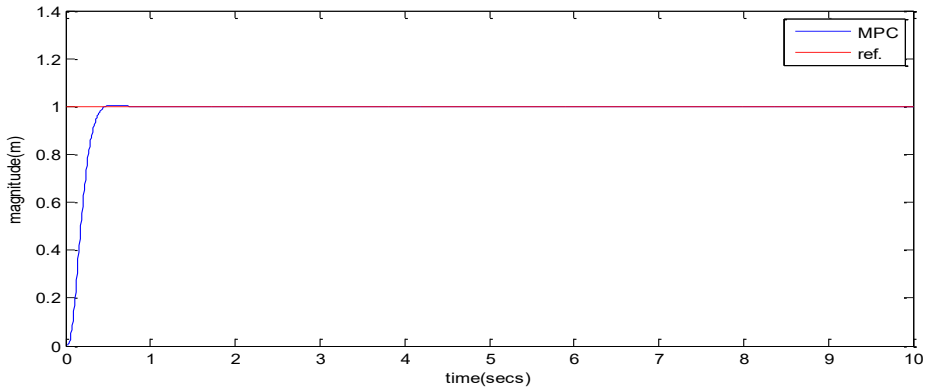


Figure 4.6 : MPC response with step input

4.3 discussions of the results

Table 4.4: Comparison table

PARAMETERS	PID	MPC
Rise time (s)	0.0284	0.2374
Settling time (s)	1.2646	0.4051
Peak time (s)	0.0925	0.5568
Maximum overshoot (%)	74.300	0.5519

From table 4.4, the two most important specifications namely: settling time and overshoot are more improved in MPC controller than in PID controller. This show

MPC take more control action hence least steady state error and more stability is achieved.

Similarly, Rise time is much improved in PID than in MPC. This shows MPC is slower to PID due to many calculations involved in its control action

Finally, the design of MPC using SIMULINK is easy and effective. Some insight knowledge of MATLAB tool box is required. The tuning of the MPC parameters also requires some time.

4.4 summary

The magnetic suspension system was simulated to obtain an open-loop response. MPC and PID controllers close loop responses were also obtained. Based on comparison of the controller's response of some transient and steady state specification, some discussion have been made.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This chapter summarises the general conclusion of the project as well as recommendation for future work.

5.2 Conclusion

This project has presented a simulation and control of magnetic suspension system as well as the comparison between the performance of MPC controller and PID controller on the system. Simulation results showed that MPC controller is the best alternative for control technique.

5.3 Recommendations

Intelligent controllers such as pole placement, LQR and Fuzzy Logic control scheme should be tested on the system to improve the rise time.

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